

Ex. No. : 05

Date:

Register No : 231701007

Name: Badrish R S

A python program to implement multi layer perceptron with back propagation.

AIM:

To implement the **Multi Layer Perceptron (MLP) with Backpropagation** using Python.

REQUIREMENTS:

Jupyter Notebook with Python.

PROGRAM:

```
# Import libraries import pandas as pd import matplotlib.pyplot
as plt from sklearn.model_selection import train_test_split from
sklearn.preprocessing import LabelEncoder, StandardScaler from
sklearn.neural_network import MLPClassifier from
sklearn.metrics import accuracy_score

# Load dataset data =
pd.read_excel("iris_dataset.xlsx") X =
data.iloc[:,1:5] y = data.iloc[:,5] encoder
= LabelEncoder() y =
encoder.fit_transform(y)
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

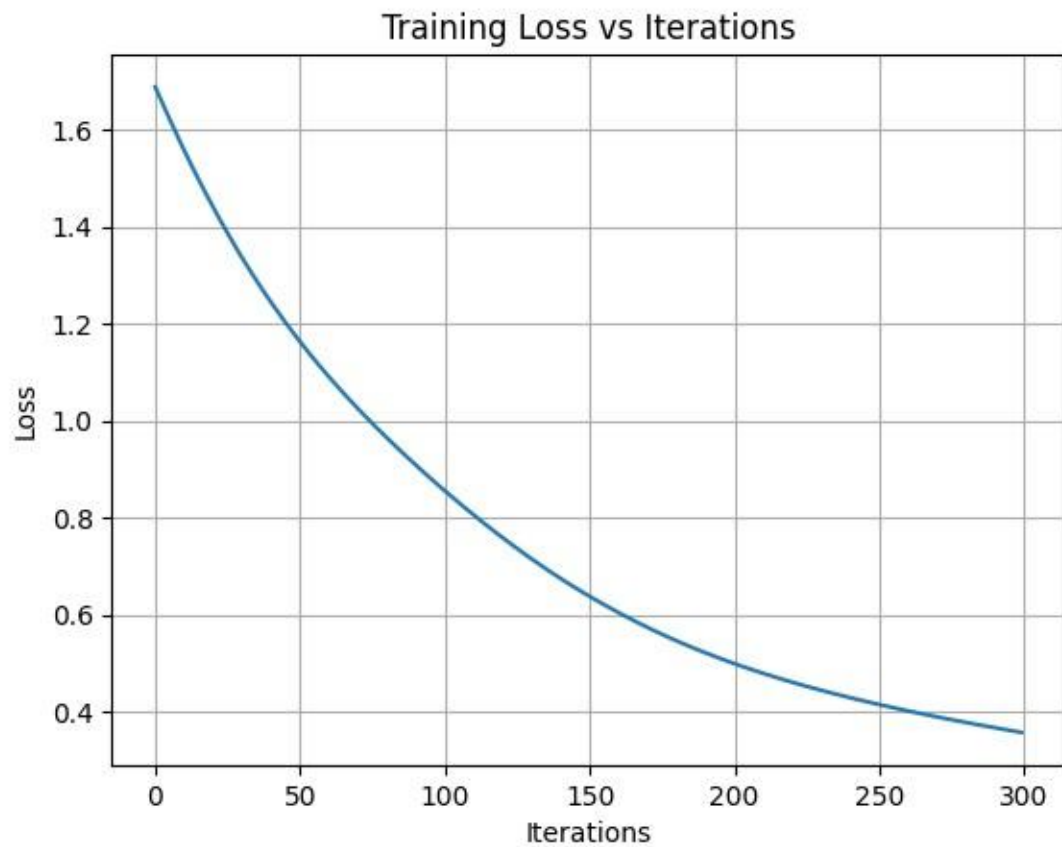
```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42)
```

```
# MLP Model mlp =  
MLPClassifier(hidden_layer_sizes=(8,8),  
              activation='relu',  
              solver='adam',          max_iter=300,  
              random_state=1) mlp.fit(X_train,  
y_train) y_pred = mlp.predict(X_test)  
accuracy = accuracy_score(y_test, y_pred)  
print("Accuracy:", accuracy)
```

```
plt.plot(mlp.loss_curve_)  
plt.title("Training Loss vs Iterations")  
plt.xlabel("Iterations")  
plt.ylabel("Loss") plt.grid(True)  
plt.show()
```

OUTPUT:

Accuracy: 0.9333333333333333



RESULT:

Thus, the Simple Linear Regression model was successfully implemented using the Least Squares Method.